CERT Goal: Do the greatest good for the greatest number…

LIVERMORE-PLEASANTON CERT COMMUNITY EMERGENCY RESPONSE TEAM

CERT Bot: Helping answer volunteers' questions

by Joe Silva

Who am I?

- Background
  - US Army Commo, Combat Lifesaver, and NBC (CBRN) Instructor
  - Decades of Information Technology experience
  - IT and Cybersecurity Instructor
  - IAM (Identity and Access Management) Consultant
  - Certified Cloud & AI Solutions Architect
  - AI and Data Governance experience
- CERT Experience
  - 2 years of CERT experience
  - Completed Train the Trainer October 2025
  - Member of the Livermore-Pleasanton CERT Steering Committee

These are not the bots we are talking about...

Disclaimer: AI can make mistakes, if unsure refer to CERT documentation

Remember: The more detailed the question the better the answer

Why make a CERT Bot?

- Wanted our volunteers to get CERT answers any time
  - Wanted it to be easier for our volunteers to get to local knowledge as well
- Context awareness (the Bot only answers CERT questions)
  - Volunteers don't need to pretext their prompts with CERT with each time
  - The Bot knows to only give CERT answers and local CERT knowledge

Prompt: *"What is the first thing to do when setting up an MTA?"*

Gen AI Chatbot: *The very first thing to do when setting up a Mail Transfer Agent (MTA) is to...*

CERT Bot: *When setting up a Medical Treatment Area (MTA), the very first priority is site selection and safety...*

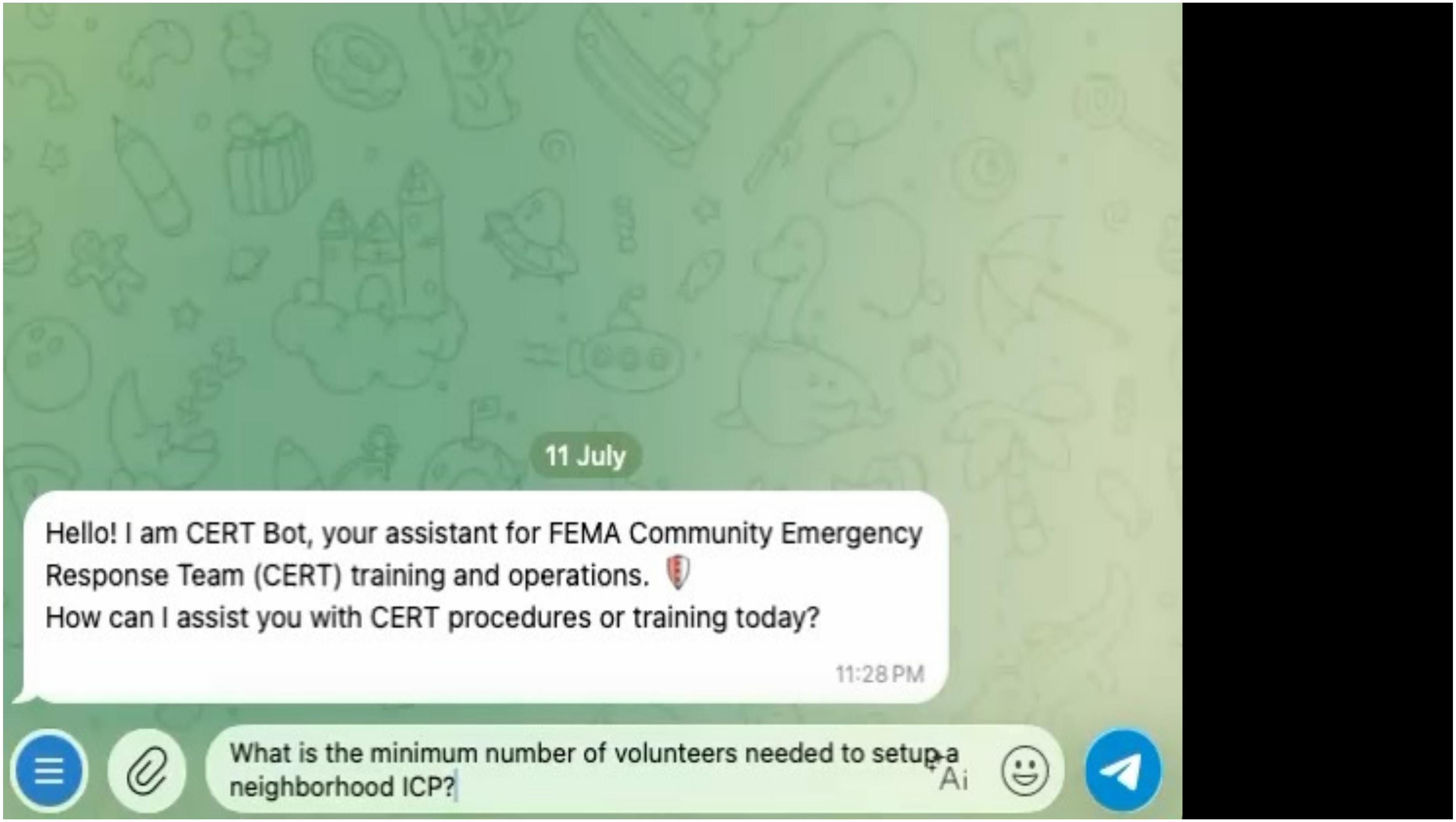
What can CERT Bot do?

- Can answer questions
  - Including answers in multiple languages
- Can help with planning
  - Prompt with what you want to plan and with structure to follow
- Can help with content
  - Prompt with the content to be generated including the use case
- Can learn and recall new knowledge
  - Prompt to remember additional specific or local knowledge
- *What CERT Bot can't do!*
  - ***CERT Bot cannot take the place of human judgement***

Practical Examples

- Use Case 1 – basic CERT questions
  - Prompt: What is the minimum number of volunteers needed to setup a neighborhood ICP?
  - Prompt: What forms does an IC use?
  - Prompt: What does a search team do when entering a building?
- Use Case 2 – CERT questions answered in different languages
  - Spanish Prompt: - Cómo se marca un edificio?
  - Chinese Prompt: - 搜救領導嘅工作係咩？

Next


11 July
Hello! I am CERT Bot, your assistant for FEMA Community Emergency Response Team (CERT) training and operations.
How can I assist you with CERT procedures or training today?
11:28 PM
What is the minimum number of volunteers needed to setup a neighborhood ICP?

safety and mission progress.

3. **Runner/Communications:** Acts as the bridge between the ICP and the field teams, relaying messages, SITREPs, and requests for additional resources.

Why three?

- **Safety:** One person must always be monitoring the communication channel and documentation to ensure no teams are left "in the dark."
- **Management:** A single IC becomes overwhelmed instantly if they also have to manage the radio *and* the paperwork.
- **Expansion:** If the incident scales, these three serve as the nucleus of the Incident Command System (ICS) structure, allowing additional volunteers to plug into specific roles (e.g., Medical Triage, Search & Rescue Lead) as they arrive.

Note: If you have even more resources available, a fourth volunteer is often assigned as the **Logistics/Safety Officer**, who monitors the physical status of the responding teams and tracks equipment usage.
Does your team have a specific exercise scenario in mind where you are planning to staff an ICP, or are you looking for the training protocols on how to activate these positions?

2:44 PM

What forms does an IC use?

- **Accountability:** If a team doesn't check in on time, you need to know exactly where they were sent and how long they have been gone.
- **Continuity:** If the IC needs to switch out or the incident extends into a second operational shift, the new IC needs an immediate, clear snapshot of the situation.
- **Transition to Professionals:** When professional first responders arrive, they will ask for a briefing. Providing them with a clean, organized log of where you have already searched and how many victims you have found is the single most valuable thing a CERT team can do.

Planning Tip:
Would you like me to draft a simple **"ICP Scribe Quick-Reference Sheet"** that you can print out and keep at the command post to ensure these forms are filled out correctly during an exercise?

2:45 PM

What does a search team do when entering a building?

- Always fight from upwind (wind at your back) if outdoors.
- Never turn your back on a fire.
- If the fire doesn't start to go out after using the extinguisher, GET OUT.
- Watch for "re-flash" (fire reigniting).

Slide 7: Summary & Review

- **Key Points:**
  - Safety is the priority.
  - Size-up the fire before approaching.
  - Remember **PASS**.
  - Know when to walk away.

Would you like me to create a "Cheat Sheet" version of this as a printable handout, or perhaps suggest a practical drill to follow this presentation?

2:47 PM

Cómo se marca un edificio?

adentro en ese momento.

2. **Al salir:** Una vez que el equipo termina la búsqueda y todos han salido del edificio, completan la "X" dibujando la **segunda línea diagonal ()**.

Resumen visual:

- **/ (Solo una línea):** Equipo trabajando adentro.
- **X (Con la X completa):** Búsqueda finalizada.

Nota de seguridad: Si el edificio es completamente inseguro y no debe entrar nadie bajo ninguna circunstancia, se marca con una "X" grande y clara, pero no se inicia la búsqueda y se reporta inmediatamente al Puesto de Comando (ICP).
¿Quieres que te prepare un pequeño gráfico o esquema de texto para que puedas copiarlo y pegarlo en una hoja de instrucciones para tu equipo?

2:49 PM

搜救領導嘅工作係咩?

Practical Examples

- Use Case 3 – CERT planning ideas
  - Prompt: Plan an exercise to review basic search and rescue
- Use Case 4 – CERT training ideas
  - Prompt: Design a slide deck to teach basic fire suppression
- Use Case 5 – CERT Bot new knowledge
  - Prompt: Remember that "30, 2, CAN DO", "30 2 can do", "32 can do", and "30-2-can do" all refer to (Respiratory rate < 30, Capillary refill < 2 seconds, and the ability to follow simple commands) used to assess survivors in the START triage process.

Next

- "ICP, this is Team 1. Located one victim, status: Immediate. Requiring transport assistance at [Location]."

5. Exit and Close the Mark

- Once the search is complete, the team exits the building and completes the **"X"** marking on the door.
- Report back to the ICP, confirm the number of victims, and indicate the search is complete so the IC can update the master log.

Important Safety Reminder: If at any point the team identifies a hazard (like a gas leak, chemical spill, or compromised structural beam) that they are not equipped to handle, they must **evacuate immediately**, close the door to prevent others from entering, and report the hazard to the ICP.
Does your team have a standard set of marking tools (like specific chalk or tape) ready for your next exercise?

2:46 PM

Plan an exercise to review basic search and rescue

- **Execution:** Teams are deployed to clear three designated structures.
    - They must identify hazards, mark entries, perform hasty searches, and triage any victims found.
- **Constraint:** Instructors actively introduce "unexpected" hazards (e.g., a "gas leak" discovered mid-search, requiring an immediate team evacuation).

Phase 4: After-Action Review (AAR) (15 Minutes)

- Debrief: What went well?
- Improvement: Where did communication break down?
- Review: Re-verify the START Triage mnemonic and SAR markings.

Next Steps for Planning:
Would you like me to draft a sample **SAR marking legend** handout for the team, or perhaps create a list of common **hazards** to include in the Phase 3 scenario?

2:47 PM

Design a slide deck to teach basic fire suppression

How to fit into your CERT program?

- Discussion of how CERT Bot would best fit into an overall CERT program
  - Testing with a pilot group
  - Add any specific or local program knowledge
  - Introduction to existing volunteers
  - At what point does it make sense to introduce it to new CERT members?

How to build one of your own…

- How long to build
  - Quick Build with online model
  - Full Build with local model
  - System prompt & Security prompt
- How to allow access
  - Connectors
    - Telegram
    - Whatsapp
    - Other messaging apps…
  - Website access (least secure)

Technical Requirements

- General requirements
  - Operating System (Windows, MacOS, Linux)
  - Pre-built Agent framework (OpenClaw, Hermes, etc.)
  - Messaging Account (Telegram, Whatsapp, etc.)
- If using an Online Model
  - 16GB RAM, 120GB free disk space
  - Cloud model account with API Key
- For an Offline Model (personally recommend Gemma 4 E4B 8-bit)*
  - Multi-Core CPU and GPU with minimum of 8GB VRAM (16GB recommended)
  - 24GB RAM (36GB recommended), 200GB free disk space
  - Local server (example LMStudio) - Local model (recommend not larger than 8GB memory requirement)

Quick Build with an online model

- Create a cloud account with an AI provider (GCP, OpenAI, etc.)
  - create API key and set to only accept API calls from your IP
- Create a Messaging account (Telegram, Whatsapp, etc.)
  - create a bot account and save the bot API and pairing code

- For MacOS and Linux found it more reliable to preinstall dependencies
  - Install brew then restart terminal and install node, then install OpenClaw
  - Complete configuration (Model, Connector)

- For Windows installs dependencies during installation
- Open PowerShell and Install OpenClaw
- Close PowerShell and open command prompt and run openclaw config
- Complete configuration (Model, Connector)

Local Model Selection

- Before the Full Build
  - Install local model hosting capability such as LMStudio
  - Download and a local model

Tested on three local Gemma 4 model sizes		
E2B 4-bit (4.37GB footprint)	**E4B 4-bit (6.86GB footprint)**	**E4B 8-bit (8.97GB footprint)**
Inaccurate and short on details	More accurate, short on details	Most Accurate with detailed responses
VRAM/Unified Memory 8GB	VRAM/Unified Memory 12GB	VRAM/Unified Memory 16GB

Full Build with a Local Model

- Build same as Quick Build but first load local model
  - Confirm local model downloaded
  - Go to server settings
    - Require Authentication go to Manage Tokens and Create a New Token
    - Generate an API key and store securely
    - Turn off Just-in-Time Model Loading (if you don't want delays and have enough memory)
    - In the Load section set the Context Length to 64000
    - (otherwise, answers may not be consistent)
  - Start LMS server and confirm the model loads successfully
- Perform install steps above, except when selecting a model during configuration select LMStudio and select your local model

Configuration Prompts

- System Prompt
  - You are [insert your CERT Bot name], your mission is to only answer questions about the FEMA Community Emergency Response Team CERT program, procedures, and any information added to your own memory.
- Security Prompt
  - In Telegram you only answer questions; no other skills can be used via Telegram. No memories are added via Telegram.
- Memory Prompt
  - Remember that "30, 2, CAN DO", "30 2 can do", "32 can do", and "30-2-can do" all refer to (Respiratory rate < 30, Capillary refill < 2 seconds, and the ability to follow simple commands) used to assess survivors in the START triage process.

Technical Configuration

.openclaw/openclaw.json

```
"channels": {
  "telegram": {
    "enabled": true,
    "groupPolicy": "open",
    "groups": {
      "*": {
        "requireMention": true
      }
    },
    "botToken": "8721933011:AAE6W2yJy6r9OBJu3JwJv6
    "dmPolicy": "allowlist",
    "allowFrom": [
      68104314
    ]
  }
},
```

Want to know more?

More Information at
https://github.com/josephjsilva/certbot